

MARCO AMATO

Postdoctoral Researcher - Solid & Structural Mechanics

RESEARCH INTERESTS

My research interests are primarily situated within the field of solid and structural mechanics with a focus on the complexities that arise when nonlinearity is involved. Within this area, I have concentrated my efforts on exploring configurational forces and the interactions between magnetic and elastic phenomena commonly referred to as magnetoelasticity. My work involves both the development of analytical models and the application of numerical simulations to verify these models, aiming to deepen the understanding of material responses and instabilities under multiphysics loading.



EDUCATION

Ph.D. - Solid & Structural Mechanics

University of Trento [📍](#) (Trento, Italy)

Nov 20 - May 24

Thesis: *Elastic solids under frictionless rigid contact and configurational force* [📄](#)

Supervisors: Prof. Francesco Dal Corso [📧](#), Prof. Davide Bigoni [📧](#) and Prof. Andrea Piccolroaz [📧](#)

M.Sc. - Civil Engineering

University of Bologna [📍](#) (Bologna, Italy)

Sep 16 - Jul 20

Thesis: *The Galerkin method for vibrational problem on stepped structures* [📄](#)

Supervisors: Prof. Alessandro Marzani [📧](#) and Prof. Isaac Elishakoff [📧](#)

B.Sc. - Civil Engineering

University of Bologna [📍](#) (Bologna, Italy)

Sep 12 - Oct 16

Thesis: *Experimental shear test on masonry samples: investigation on the role of prestress* (in Italian) [📄](#)

Supervisors: Prof. Cristina Gentilini [📧](#), Prof. Stefano de Miranda [📧](#) and Dr. Antonio Maria D'Altri [📧](#)

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

Faculty of Civil Engineering [📍](#) - Department of Mechanics [📍](#)

Czech Technical University in Prague [📍](#) (Prague, Czech Republic)

Feb 24 - Present

Project: **SOFFA ERC CZ** [📄](#)

I focus on *magnetorheological elastomers* (MREs), a class of smart materials that combine elastomers with magnetic particles, allowing their mechanical properties to be tuned by an external magnetic field. My work involves developing variationally consistent models to describe the behavior of these materials under various mechanical and magnetic conditions, including large deformations and instability phenomena. These models are then implemented into OOFEM [📄](#) an open-source finite element software, written in C++ and developed within the department, enabling more accurate simulations of MREs in practical applications.



Project: **Beyond ERC-AdG** [↗](#)

My research focused on *configurational forces* in solids. Specifically, I analyzed the behavior of a hyperelastic block in frictionless contact with a rigid punch. Under partial compression, a configurational force arises at the punch's corner, encouraging the material to be ejected from the compressed zone. This phenomenon was investigated analytically and subsequently validated through numerical simulations.

Visiting Scholar

Aug 19 - Mar 20

Department of Ocean & Mechanical Engineering [↗](#)
Florida Atlantic University [↗](#) (Boca Raton, Florida, US)

Host & Supervisor: Prof. Isaac Elishakoff [↗](#)

Research internship for Master's thesis devoted to the development of the Galerkin method for stepped structures. I compared standard and generalized functions approaches by applying them to various structural elements like bars, beams and plates.

Lab Intern

Jun 16 - Feb 17

Department of Civil, Chemical, Environmental and Materials Engineering [↗](#)
University of Bologna [↗](#) (Bologna, Italy)

Experimental research for Bachelor's thesis. Experiments involved the assessment of the elastic modulus of mortar and brick materials, measurement of the flexural strength of mortar specimens, the development of pull-out tests on steel helical bars and the evaluation of the shear strength of masonry specimens.

TEACHING

Invited Lecturer

- **An Introduction to Nonlinear Solid Mechanics**
University of Palermo [↗](#) (Palermo, Italy)
AY 2025/26 | PhD course

Teaching Assistant

- **Scienza delle Costruzioni** (Solid Mechanics)
University of Trento [↗](#) (Trento, Italy)
AY 2021/22, 2022/23, 2023/24 | First semester | Undergraduate course
- **Meccanica Computazionale delle Strutture 2** (Computational Mechanics 2)
University of Trento [↗](#) (Trento, Italy)
AY 2021/22 | Second semester | Graduate course
- **Teoria e Dinamica delle Strutture** (Theory and Dynamics of Structures)
University of Trento [↗](#) (Trento, Italy)
AY 2020/21, 2021/22, 2022/23 | First semester | Graduate course

Guest Lecturer

- **Mechanical Vibrations**
Florida Atlantic University [↗](#) (Boca Raton, Florida, US)
AY 2019/20 | First semester | Graduate course
- **Statics**
Florida Atlantic University [↗](#) (Boca Raton, Florida, US)
AY 2019/20 | First semester | Undergraduate course

Additional Teaching Experience

- Individual tutoring to high school students in Mathematics and Physics to improve their understanding and prepare them for the tests.



AWARDS

Finalist GMA Award for the Best PhD Thesis in Mechanics of Materials

Apr 25

Mechanics of Materials Group (GMA) [↗](#), **AIMETA** [↗](#)

Shortlisted among the top three PhD theses in Mechanics of Materials defended in Italy in 2024, as part of a national competition organized by the Mechanics of Materials Group (GMA) of the Italian Association of Theoretical and Applied Mechanics (AIMETA).

Iniciační Fond FSv

Apr 25

Czech Technical University in Prague [↗](#) (Prague, Czech Republic)

Amount: 50 000.00 CZK (~2 000.00 EUR)

Awarded funding from the *Iniciační Fond FSv* to host Prof. Luis Dorfmann and develop analytical and numerical models for magnetoelastic interactions through international research collaboration.

Scholarship for Final Thesis Research Abroad

Jun 19

University of Bologna [↗](#) (Bologna, Italy)

Amount: 2 237.53 EUR

Awarded the scholarship under the *Scholarships for Final Thesis Research Abroad in Second-Cycle or Single-Cycle Degrees* program to support international research for a final thesis.

PUBLICATIONS

Peer-Reviewed Articles

- [5] Dal Corso, F., Amato, M., & Bigoni, D. (2024). Elastic solids under frictionless rigid contact and configurational force. *Journal of the Mechanics and Physics of Solids*, 188, 105673.
DOI: <https://doi.org/10.1016/j.jmps.2024.105673> [↗](#)
- [4] Amato, M., Elishakoff, I., & Reddy, J. N. (2021). Flutter of a Multicomponent Beam in a Supersonic Flow. *AIAA Journal*, 59(11), 4342-4353.
DOI: <https://doi.org/10.2514/1.J060631> [↗](#)
- [3] Elishakoff, I., & Amato, M. (2021). Flutter of a beam in supersonic flow: truncated version of Timoshenko-Ehrenfest equation is sufficient. *International Journal of Mechanics and Materials in Design*, 1-17.
DOI: <https://doi.org/10.1007/s10999-021-09537-x> [↗](#)
- [2] Elishakoff, I., Amato, M., & Marzani, A. (2021). Galerkin's method revisited and corrected in the problem of Jaworski and Dowell. *Mechanical Systems and Signal Processing*, 155, 107604.
DOI: <https://doi.org/10.1016/j.ymssp.2020.107604> [↗](#)
- [1] Elishakoff, I., Amato, M., Ankitha, A. P., & Marzani, A. (2020). Rigorous implementation of the Galerkin method for stepped structures needs generalized functions. *Journal of Sound and Vibration*, 115708.
DOI: <https://doi.org/10.1016/j.jsv.2020.115708> [↗](#)

Book Chapters

- [1] Bigoni, D., Amato, M., & Dal Corso, F. (2022). Configurational Forces in Penetration Processes. *In Solid (Bio) mechanics: Challenges of the Next Decade: A Book Dedicated to Professor Gerhard A. Holzapfel* (pp. 429-437). Cham: Springer International Publishing.
DOI: https://doi.org/10.1007/978-3-030-92339-6_19 [↗](#)

Conference Papers

- [1] Gentilini, C., D'Altri, A. M., Amato, M., Zanotti, P., Favaro, F., & de Miranda, S. (2017). Salt attack effects on the shear behavior of masonry: preliminary results of an experimental campaign. *In Key Engineering Materials* (Vol. 747, pp. 512-517). Trans Tech Publications Ltd.
DOI: <https://doi.org/10.4028/www.scientific.net/KEM.747.512> [↗](#)



TALKS & EVENTS

Organizing Activities

RAMSS2025 - Recent Advances in Mechanics of Soft Solids [↗](#)
Prague, Czech Republic

Jun 10, 2025

Co-organizer of this international event in collaboration with Martin Horák [↗](#).

Invited Presentations

Nečas Seminar on Continuum Mechanics [↗](#)
Prague, Czech Republic

Apr 14, 2025

Elastic solids under frictionless rigid contact and configurational force.

Contributed Talks

YIC 2025 - ECCOMAS 8th Young Investigators Conference [↗](#)
Pescara, Italy

Sep 17 - 19, 2025

Hard-magnetic soft materials: theory and implementation.

1st HICOMP Conference - 1st Hellenic-Italian Conference on Computational Mechanics, Biomechanics and Mechanics of Materials [↗](#)
Rhodes Island, Greece

Jun 19 - 21, 2025

Hard-magnetic soft materials: theory and implementation.

NMM2024 - Conference on Nano and Macro Mechanics [↗](#)
Prague, Czech Republic

Sep 12, 2024

Hard-magnetic soft materials: theory and implementation.

ICNEM - 26th International Conference on Nonlinear Elasticity in Materials [↗](#)
Prague, Czech Republic

Jun 9 - 14, 2024

Efficient numerical approach to modeling of hard magnetic soft beams.

Mathematics and Mechanics of Solids and Structures - Scientific challenges and methodologies for future societal development [↗](#)
Aberystwyth, Wales, UK

Jun 7 - 9, 2023

Elastic solids moving along frictionless constraints via configurational forces.

EUROMECH - Mechanics of high-contrast elastic composites [↗](#)
Virtual Conference, Online

Sep 6 - 8, 2021

Free vibration and flutter of a multi-component beam in a supersonic flow.

ASME - International Mechanical Engineering Congress & Exposition [↗](#)
Virtual Conference, Online

Nov 16 - 19, 2020

Flutter of a beam in a supersonic flow: no need in full Timoshenko-Ehrenfest equations.



PEER REVIEW

- Reviewer for *International Journal of Non-Linear Mechanics* [↗](#)
- Reviewer for *Computers and Structures* [↗](#)

Last Update: March 31, 2026